From Prompting to Social Cognition: Mechanistic Deepening of PAS and FAS Frameworks in Large Language Models

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Abstract

Large language models (LLMs) have recently been used to simulate human social attitudes and collective preferences. However, most existing work treats social surveys as one-shot prompting tasks, offering limited insight into the mechanisms of social cognition in LLMs. Building on the Partial Attribute Simulation (PAS) and Full Attribute Simulation (FAS) frameworks, we propose a data-driven Social Cognition Simulation Pipeline (SCSP) that turns these paradigms from performance benchmarks into tools for mechanistic analysis of social reasoning. We construct probabilistically grounded datasets by reconstructing joint distributions from real social surveys, and design a fine-tuning strategy that explicitly aligns model outputs with target marginal and conditional distributions. This enables LLMs to internalize both individual-level reasoning consistency (via PAS) and population-level distributional coherence (via FAS). Empirically, our approach improves candidate preference prediction accuracy from 13% to 60% overall (from 0% to 72% on the most challenging cases), and substantially stabilizes simulated opinion distributions, yielding over 80% reduction in Jensen–Shannon / KL divergence on FAS tasks. Notably, these gains are achieved by compact, fine-tuned models that outperform several much larger state-of-the-art LLM baselines on the same social cognition benchmarks.

1 Introduction

2 Related Works

Social Representation and Pragmatic Deficits. Despite linguistic prowess, LLMs often lack deep social comprehension. Generic models frequently exhibit biases skewed towards WEIRD populations (Santurkar et al., 2023; Ma et al., 2024), failing to reflect diverse perspectives. While prompting strategies (Moon et al., 2024) offer temporary behavioral guidance, they function as external constraints rather than fostering internalized understanding. Similarly, treating social reasoning merely as classification (Feng et al., 2023) neglects the generative mechanics essential for genuine interaction. We address this by shifting from surface-level constraints to the mechanistic deepening of Social Cognition and Social Pragmatics (SCSP).

Competence vs. Preference Alignment. Supervised Fine-Tuning (SFT) successfully embeds complex logic in physical and financial domains (Lai et al., 2024; Chang et al., 2025, 2024), demonstrating its potential to teach systemic rules. However, current social domain applications primarily utilize SFT for ideological alignment (Chu et al., 2023; Zhao et al., 2023), focusing on preference mimicry rather than cognitive reasoning. Diverging from this trend, we leverage SFT to align model *competence* with human cognitive frameworks. Our goal is to instill the "rules of social relations"—specifically Faux Pas detection and pragmatic inference—rather than biasing the model towards specific group voices.

Data-Centric Social Grounding. Model alignment fidelity fundamentally depends on data quality (Sun et al., 2025). Existing datasets, often reliant on unstructured QA, lack the depth to teach relational nuances. Building upon the **LLM-S^3 Benchmark** (Zhao et al., 2025)—which established the PAS and FAS evaluation frameworks—we extend these theories from *evaluation* to *training*. We introduce a structured training corpus derived and expanded from authoritative sociological surveys (ANES and GSS). By mapping these scenarios to the logical decomposition of social errors, we ensure that SCSP principles are mechanistically internalized, moving beyond mere prompting.

3 Methodology

We designed an integrated social cognition simulation pipeline (SCSP) that enhances the performance of PAS and FAS through structured data design, probabilistic alignment, and lightweight fine-tuning(Zhao et al., 2025). Unlike previous setups that relied solely on cueing, our approach instantiates two specialized models—one for PAS and one for FAS—within a shared pipeline, maintaining distributional consistency between individual and group-level simulations. Both models are based on smaller, faster Llama-3 variants, and we compare them to larger state-of-the-art baselines.

In the remainder of this section, we first describe the PAS component, where we use targeted data design and fine-tuning to improve 0–100 rating-based multiple-choice performance. We then briefly outline the distinct processing strategy used for FAS, which focuses on population-level distributional coherence.

3.1 PAS: Data Design for Rating-Based Multiple Choice

The *Partial Attribute Simulation* (PAS) component casts 0–100 thermometer ratings as a multiple-choice prediction task. Given a voter profile v and a candidate or group set $\{c_k\}$, the model predicts each target's rating and induces a choice via $\arg\max$. We aim to improve this rating-based multiple-choice accuracy through structured data design and lightweight fine-tuning of a compact Llama-3 model.

From wide table to interaction-level data. Let v denote a voter profile (e.g., demographics, party identification, ideology), c a target (candidate or social group), $y \in [0, 100]$ the ANES thermometer score, and t the survey year. In the original *wide* format, each row is a voter and each column a target:

$$\text{Voter}_i : [y_{i,\text{Biden}},\ y_{i,\text{Trump}}, \dots],$$

which implicitly models $P(y \mid v)$ and suppresses voter–target interactions. We instead construct a *long* format in which each row is a voter–target pair,

$$(v_i, c_j, y_{ij}, t_i) \quad \text{for all non-missing } y_{ij},$$

so that PAS learns $P(y \mid v, c, t)$ from roughly $N_v \times N_c$ interaction-level samples.

We apply the same transformation not only to candidate ratings but also to attitudes toward social groups (e.g., racial groups, parties, ideological groups, social movements). For the group-attitude extension, each row corresponds to a voter–group pair (e.g., voter–Black Americans, voter–White Americans), yielding an additional long-format dataset with 61,710 interaction-level samples across nine groups (four racial groups, two parties, two ideological groups, and one social movement). Both candidate-level and group-level interactions share the same voter feature space, enabling a unified PAS formulation.

Each target c is represented by a structured metadata vector

$$\mathcal{M}[c] = (\text{party}_c,\ \text{ideo}_c,\ \text{age}_c,\ \text{incumbent}_c, \dots),$$

for candidates, and an analogous set of fields (e.g., category, ideological position, political leaning, valence) for social groups. PAS predicts ratings as

$$\hat{y} = f_\theta^{\text{PAS}}(v, \mathcal{M}[c], t),$$

where θ are LoRA-adapted Llama-3 parameters. At evaluation time, given v and a candidate set $\{c_1, \dots, c_K\}$, the model outputs $\hat{y}_k$ and selects

$$\hat{c}^* = \arg\max_k \hat{y}_k,$$

which we compare to the observed vote choice. For group-attitude prediction, we evaluate PAS on rating error and custom accuracy over voter–group pairs.

Social-cognitive interaction features. On top of the long-format data, we add a minimal set of interaction features that encode core social-cognitive regularities.

Partisan ingroup versus outgroup is encoded as

$$\text{party_match}(v, c) = \begin{cases} 1, & \text{if party}_v = \text{party}_c, \\ 0, & \text{otherwise.} \end{cases}$$

Ideological proximity on the 7-point scale is captured by

$$\text{ideo_dist}(v, c) = |\text{ideo}_v - \text{ideo}_c|,$$

with an approximately linear spatial pattern

$$\mathbb{E}[y \mid \text{ideo_dist} = d] \approx \beta_0 - \beta_1 d, \quad \beta_1 > 0.$$

To combine party and ideology, we use a compact similarity index

$$\text{sim}(v, c) = 0.5\,\mathbb{1}[\text{party_match} = 1] + 0.5\left(1 - \frac{\text{ideo_dist}(v, c)}{6}\right),$$

where higher sim is associated with more favorable ratings.

We also flag cross-pressured cases as hard examples, e.g.,

$$\mathcal{CP}(v, c) = \mathbb{1}[\text{party_match} = 1 \wedge \text{ideo_dist}(v, c) \geq 3],$$

which captures co-partisans who are ideologically distant. These pairs are oversampled during training to discourage trivial "always pick co-partisans" heuristics. The same feature design and sampling strategy are applied consistently across both candidate and group targets.

Fine-tuning objective. Let $P_{\text{data}}(v, c, y)$ denote the distribution induced by the combined long-format ANES tables (candidates and groups) with interaction features. PAS is fine-tuned to minimize

$$\theta^* = \arg\min_{\theta} \mathbb{E}_{(v,c,y)\sim P_{\text{data}}}\left[\mathcal{L}(f_\theta^{\text{PAS}}(v, \mathcal{M}[c], t), y)\right]$$

so that $f_{\theta^*}^{\text{PAS}}(v, \mathcal{M}[c], t)$ approximates the conditional expectation of thermometer scores for both candidate preferences and group attitudes. Converting these scores into choices via $\arg\max$, the interaction-level, feature-augmented, and multi-target design yields substantial gains in 0–100 rating-based multiple-choice accuracy over cue-only baselines, while also improving prediction of social group attitudes such as racial and partisan evaluations.

3.2 FAS: Population-Level Full Attribute Simulation

The *Full Attribute Simulation* (FAS) component extends SCSP from individual-level judgments to population-level opinion distributions. While PAS models $P(y \mid v, c, t)$ for voter–candidate interactions, FAS focuses on constructing and sampling from a structured joint distribution (Long et al., 2013) over key categorical attributes (e.g., policy attitudes) for a target year t^*, and training a compact Llama-3 model to reproduce and manipulate these distributions in natural language.

Problem setup. Let A, B, C denote three discrete attributes of interest (e.g., attitudes toward different policy domains) with finite supports, and let $t_1, \ldots, t_n$ be historical survey years with observed empirical distributions $\{p_{t_i}\}$. For a target year t^* (e.g., a future election), FAS aims to (i) extrapolate the marginal and pairwise distributions to t^*, (ii) construct a coherent three-way joint distribution $Q^*(A, B, C)$ consistent with these predictions, and (iii) generate a diverse set of fine-tuning tasks that teach the model to generate, analyze, and adjust population-level samples.

Multi-model ensemble extrapolation. For each univariate or pairwise cell (e.g., $P(A = a)$, $P(A = a, B = b)$), we observe a time series $\{p_{t_i}\}_{i=1}^n$ and predict p_{t^*} via a four-model ensemble:

$$\hat{p}_{\text{poly}}(t^*), \quad \hat{p}_{\text{loess}}(t^*), \quad \hat{p}_{\text{exp}}(t^*), \quad \hat{p}_{\text{linear}}(t^*),$$

corresponding to quadratic polynomial regression, LOESS with tri-cube kernel, exponential trend, and linear regression, respectively. The ensemble prediction is

$$\hat{p}(t^*) = \frac{1.0\,\hat{p}_{\text{poly}} + 1.0\,\hat{p}_{\text{loess}} + 0.6\,\hat{p}_{\text{exp}} + 0.8\,\hat{p}_{\text{linear}}}{3.4},$$

with values clipped to $[0, 1]$ and renormalized. This yields predicted marginals $P^*(A), P^*(B), P^*(C)$ and pairwise joints $P^*(A, B), P^*(A, C), P^*(B, C)$ for the target year, while mitigating overfitting to sparse and noisy historical data.

Joint distribution via IPF. To recover a coherent three-way joint $Q^*(A, B, C)$ that respects all predicted marginals and pairwise tables, we apply iterative proportional fitting (IPF). Starting from the independence approximation

$$Q^{(0)}(a, b, c) = P^*(A = a)\,P^*(B = b)\,P^*(C = c),$$

we alternate multiplicative updates to match each pairwise table:

$$Q(a, b, c) \leftarrow Q(a, b, c) \cdot \frac{P^*(B = b, C = c)}{\sum_{a'} Q(a', b, c) + \epsilon},$$

and similarly for $P^*(A, B)$ and $P^*(A, C)$, followed by updates to match the three marginals $P^*(A), P^*(B), P^*(C)$. After a fixed number of iterations (e.g., 800), we normalize and apply Laplace smoothing,

$$Q_{\text{final}}(a, b, c) = \frac{Q(a, b, c) + \alpha}{\sum_{a', b', c'}(Q(a', b', c') + \alpha)},$$

with small ϵ, α for numerical stability. Under standard conditions IPF converges to the unique distribution minimizing KL divergence to the initializer while satisfying all marginal and pairwise constraints, thereby preserving key inter-attribute correlations.

Task construction for population simulation. Given $Q^*(A, B, C)$, FAS samples synthetic populations and converts them into a mixture of five instruction-following task types used to fine-tune a compact Llama-3 model: (A) batch generation of respondent samples from the target distribution; (B) prior-constrained sampling, where the model must generate samples matching specified marginal targets; (C) scenario variation, where distributions are perturbed to reflect hypothetical policy or temporal scenarios; (D) subgroup analysis, where the model estimates conditional probabilities and confidence intervals (e.g., using Wilson intervals); and (E) iterative correction, where the model adjusts previously generated samples to reduce deviations from target distributions. Task proportions (e.g., 50%/25%/15%/7%/3% for A–E) are chosen to balance generation, analysis, and feedback capabilities.

To increase robustness, priors and scenarios are drawn from mild perturbations of Q^* (e.g., $\pm 10\%$ around true marginals) mixed with randomly sampled priors, and dynamic random seeds ensure that repeated runs from the same configuration yield diverse fine-tuning corpora.

FAS within SCSP. Within SCSP, FAS provides a population-level counterpart to PAS. PAS supplies micro-level behavioral patterns (e.g., how individuals with certain attributes evaluate candidates), while FAS supplies a realistic joint distribution over those attributes for a given year or scenario. Combining PAS-conditioned behavior with FAS-generated synthetic populations allows SCSP to simulate election outcomes and opinion distributions that are both behaviorally grounded and distributionally consistent with extrapolated survey data.

3.3 Evaluation Metric

We use complementary evaluation metrics to assess the individual-level PAS component and the population-level FAS component within SCSP.

PAS: Individual-level prediction accuracy. PAS is evaluated on the 0–100 thermometer rating task and the induced multiple-choice decision.

For rating prediction, we report mean absolute error (MAE) and root mean squared error (RMSE) over voter–target pairs:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|,$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2},$$

where y_i and $\hat{y}_i$ are the true and predicted thermometer scores, respectively.

For the rating-based multiple-choice decision, we convert scores into a predicted choice via

$$\hat{c}^*(v) = \arg\max_k \hat{y}_k(v),$$

and compute top-1 accuracy (ACC) against the observed choice:

$$\mathrm{ACC} = \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left[\hat{c}_i^* = c_i^{\text{true}}\right].$$

MAE and RMSE capture continuous calibration of social evaluations, while ACC measures discrete decision quality in rating-based multiple-choice settings.

FAS: Distributional alignment at the population level. FAS is evaluated by comparing simulated opinion distributions with empirical survey distributions. Let P denote the ground-truth distribution (from ANES or GSS) and $\hat{P}$ the distribution induced by FAS-generated samples.

For marginal and low-dimensional distributions (e.g., single attitudes or small attribute subsets), we use Jensen–Shannon divergence (JS)(Menéndez et al., 1997) and Kullback–Leibler divergence (KL):

$$\mathrm{JS}(P \,\|\, \hat{P}) = \tfrac{1}{2}\mathrm{KL}(P \,\|\, M) + \tfrac{1}{2}\mathrm{KL}\left(\hat{P} \,\middle\|\, M\right),$$

where $M = \frac{1}{2}(P + \hat{P})$,

$$\mathrm{KL}(P \,\|\, \hat{P}) = \sum_x P(x)\log\frac{P(x)}{\hat{P}(x)}.$$

To assess whether FAS preserves higher-order structure, we additionally compute JS and KL on the full joint distribution over key attributes (e.g., a three-way joint over attitudes) obtained from empirical IPF reconstructions and FAS samples. We denote these metrics as joint JS and joint KL, respectively. Lower JS/KL and joint JS/KL indicate better population-level alignment, capturing not only marginal frequencies but also inter-attribute dependencies critical for coherent social opinion simulation.

4 Experiments

experiment overview

4.1 PAS Experiments

We empirically evaluate PAS on ANES-based rating prediction and rating-induced choice, covering both candidate preferences and group attitudes. All models are built on Llama-3.1-8B-Instruct with LoRA adaptation.

Experimental setup. For candidate evaluation, we use the long-format ANES voter–candidate dataset described in Section 3.1, where each instance encodes a voter profile, candidate metadata, and interaction features (party match, ideological distance, similarity, cross-pressure). For group evaluation, we construct an analogous long-format voter–group dataset over nine social targets (four racial groups, two parties, two ideological groups, one social movement), yielding 61,710 voter–group pairs. In both cases, the target is a 0–100 feeling thermometer score.

We fine-tune a single PAS model on the union of candidate and group samples, using instruction-style prompts that describe (i) survey background and task setting, (ii) respondent profile, (iii) candidate or group attributes (including party/ideology metadata for groups where applicable), and (iv) task demands and answer format. Training uses standard causal LM loss over the numeric target token. We adopt LoRA with rank $r = 8$, $\alpha = 32$, dropout 0.1, applied to attention and MLP projections, and train for 3 epochs with a learning rate of 10^{-4}, effective batch size 16, and max sequence length 512.

Few-shot prompting. To bridge training and evaluation, we explicitly encode the PAS task as few-shot regression in the prompt. For candidate preference, the system message defines the task as predicting 0–100 support scores from demographics and political behavior; user messages present a new voter–candidate pair; and the assistant must respond with only a single integer. For group attitudes, we extend the system message to social groups and include a compact set of in-context exemplars that demonstrate how party alignment, ideological distance, and group type (e.g., "Black Americans", "Republican Party", "Socialists") map to thermometer scores. These few-shot exemplars are fixed across the test set and serve as a lightweight, interpretable prior on social-cognitive patterns, while the fine-tuned weights capture the finer-grained, distributional structure.

Baselines and evaluation. We compare the fine-tuned PAS model against (i) the untuned Llama-3.1-8B-Instruct with the same prompts (zero-shot + few-shot), and (ii) cue-only heuristics based on party match and ideological proximity (e.g., "always pick co-partisans", or ranking candidates by similarity index). For thermometer prediction, we compute mean absolute error (MAE) and root mean squared error (RMSE) over all voter–target pairs. For decision quality, we discretize predicted scores and evaluate (a) 5-bin rating accuracy (strongly unfavorable / unfavorable / neutral / favorable / strongly favorable) for individual targets, and (b) top-1 choice accuracy when selecting among competing candidates via $\arg\max$ over predicted scores. For group attitudes, we additionally report a relaxed accuracy that counts a prediction as correct if it falls in the same 20-point bin as the ground truth or differs by at most 10 points, and we analyze MAE stratified by group type and by cross-pressure indicators.

Results. As shown in Table (candidates) and Table (groups), PAS fine-tuning substantially improves both continuous calibration and discrete decisions. On candidate choice, accuracy increases from around 13% under cue-only heuristics and untuned prompting to over 60% with PAS, and from 0% to 72% on the most difficult cross-pressured subsets. MAE and RMSE likewise decrease markedly on thermometer prediction. On group attitudes, the fine-tuned PAS model achieves consistent gains in MAE, RMSE, and relaxed accuracy across all nine groups, while preserving meaningful differences in warmth across racial, partisan, and ideological targets. These results indicate that interaction-level fine-tuning on real survey-derived data, combined with structured prompts and few-shot exemplars, enables a compact Llama-3.1 model to internalize both candidate-level preferences and group-level attitudes in a socially coherent way, substantially outperforming larger, purely prompted LLM baselines on the same PAS tasks. The first line of the file must be

```
\documentclass[11pt]{article}
```

To load the style file in the review version:

```
\usepackage[review]{acl}
```

For the final version, omit the `review` option:

Command	Output	Command	Output
{\"a}	ä	{\c c}	ç
{\^e}	ê	{\u g}	ğ
{\`i}	ì	{\l}	ł
{\.I}	İ	{\~n}	ñ
{\o}	ø	{\H o}	ő
{\'u}	ú	{\v r}	ř
{\aa}	å	{\ss}	ß

Table 1: Example commands for accented characters, to be used in, *e.g.*, BibTEX entries.

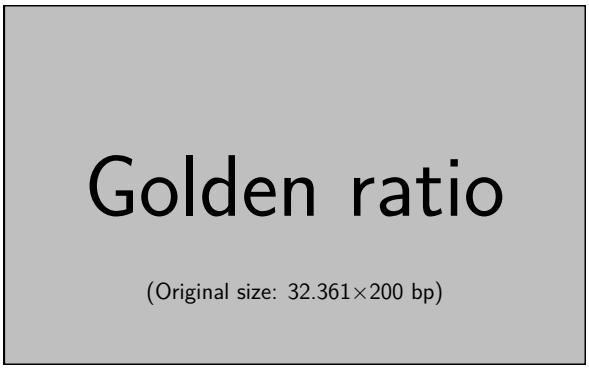


Figure 1: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

```
\usepackage{acl}
```

To use Times Roman, put the following in the preamble:

```
\usepackage{times}
```

(Alternatives like txfonts or newtx are also acceptable.)

Please see the LATEX source of this document for comments on other packages that may be useful.

Set the title and author using `\title` and `\author`. Within the author list, format multiple authors using `\and` and `\And` and `\AND`; please see the LATEX source for examples.

By default, the box containing the title and author names is set to the minimum of 5 cm. If you need more space, include the following in the preamble:

```
\setlength\titlebox{<dim>}
```

where `<dim>` is replaced with a length. Do not set this length smaller than 5 cm.

5 Document Body

5.1 Footnotes

Footnotes are inserted with the `\footnote` command.[1]

5.2 Tables and figures

See Table 1 for an example of a table and its caption.
Do not override the default caption sizes.

As much as possible, fonts in figures should conform to the document fonts. See Figure 1 for an example of a figure and its caption.

Using the `graphicx` package graphics files can be included within figure environment at an appropriate point within the text. The `graphicx` package supports various optional arguments to control the appearance of the figure. You must include it explicitly in the LATEX preamble (after the `\documentclass` declaration and before `\begin{document}`) using `\usepackage{graphicx}`.

5.3 Hyperlinks

Users of older versions of LATEX may encounter the following error during compilation:

> `\pdfendlink` ended up in different nesting level than `\pdfstartlink`.

This happens when pdfLATEX is used and a citation splits across a page boundary. The best way to fix this is to upgrade LATEX to 2018-12-01 or later.

5.4 Citations

Table 2 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get "author (year)" citations, like this citation to a paper by Gusfield (1997). You can use the command `\citep` (cite in parentheses) to get "(author, year)" citations (Gusfield, 1997). You can use the command `\citealp` (alternative cite without parentheses) to get "author, year" citations, which is useful for using citations within parentheses (e.g. Gusfield, 1997).

A possessive citation can be made with the command `\citeposs`. This is not a standard natbib command, so it is generally not compatible with other style files.

5.5 References

The LATEX and BibTEX style files provided roughly follow the American Psychological Association

[1] This is a footnote.

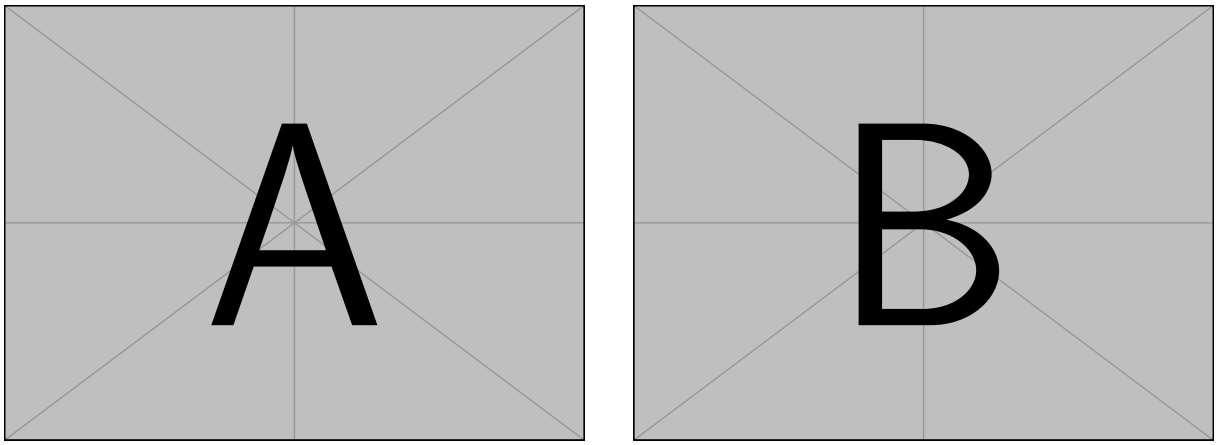


Figure 2: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(Gusfield, 1997)	\citep	
Gusfield, 1997	\citealp	
Gusfield (1997)	\citet	
(1997)	\citeyearpar	
Gusfield's (1997)		\citeposs

Table 2: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

format. If your own bib file is named `custom.bib`, then placing the following before any appendices in your LaTeX file will generate the references section for you:

```
\bibliography{custom}
```

You can obtain the complete ACL Anthology as a BibTeX file from https://aclweb.org/anthology/anthology.bib.gz. To include both the Anthology and your own .bib file, use the following instead of the above.

```
\bibliography{anthology,custom}
```

Please see Section 6 for information on preparing BibTeX files.

5.6 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command.

This an example cross-reference to Equation 1.

5.7 Appendices

Use `\appendix` before any appendix section to switch the section numbering over to letters. See Appendix A for an example.

6 BibTeX Files

Unicode cannot be used in BibTeX entries, and some ways of typing special characters can disrupt BibTeX's alphabetization. The recommended way of typing special characters is shown in Table 1.

Please ensure that BibTeX records contain DOIs or URLs when possible, and for all the ACL materials that you reference. Use the `doi` field for DOIs and the `url` field for URLs. If a BibTeX entry has a URL or DOI field, the paper title in the references section will appear as a hyperlink to the paper, using the hyperref LaTeX package.

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A Example Appendix

This is an appendix.